

Hansi Udapola

UX Researcher | XR-AI Interaction

5/89 King Street, Sydenham, Christchurch 8023, New Zealand

📞 (+64) 220 9 220 25 | ✉️ hansiudapola2@gmail.com | 🌐 hansi-udapola | 🐙 HansiUdapola

“Believe you can, and you’re halfway there.” - Theodore Roosevelt

SUMMARY

Passionate UX Engineer and Postdoctoral Researcher with expertise in designing and developing interactive systems, particularly using Unity 3D. Experienced in building cross-platform applications (WebGL, Meta, Android, Windows/macOS) and applying gamification and serious games through PhD research. Skilled in integrating conversational AI within immersive and health-related environments. Combines strong technical development capabilities with academic experience in tertiary education, bringing both practical and pedagogical insight to digital learning solutions.

WORK EXPERIENCE

Jix Reality pvt. Ltd.

Christchurch, New Zealand

Postdoctoral Researcher/Software Developer (R&D)

Oct.2025 - present

- **AI & XR Software Development:** Built interactive XR prototypes integrating AI models and real-time interactions.
- **Proof-of-Concept Engineering:** Rapid prototyping to validate research ideas into working demos and testable solutions.
- **Client Discovery & Empathy:** Client meetings, stakeholder interviews, requirements gathering, and insight translation.
- **UX & Interaction Design (XR):** Implemented user-centered UX, interaction design, and seamless AI-driven experiences.
- **Mentorship & Technical Leadership:** Mentored interns, provided technical guidance, and delivered milestones to deadlines.
- **Emerging Tech R&D:** Evaluated new frameworks, tools, and AI/XR platforms to support innovation.

Jix Reality pvt. Ltd.

Christchurch, New Zealand

HCI Research Fellow

Nov.2024 - Feb.2025

- **Opportunity Discovery & Market Insight:** Web-scraped and analysed Unity asset market data to identify gaps/opportunities; delivered actionable recommendations.
- **UCD Research Planning:** Prepared UCD user study designs, including research plans and study protocols.
- **Insight Synthesis (Human-AI Interaction):** Conducted literature review and produced human-AI virtual interaction design guidelines.
- **Feasibility Study Documentation (Digital Humans):** Documented feasibility study reporting on digital human experience considerations.
- **R&D Knowledge Documentation:** Consolidated past R&D project outputs into clear, reusable records.

Victoria University of Wellington

Wellington, New Zealand

Academic Tutor - UX Engineering

Feb.2023 - Jun.2025

- **UX Research & Evaluation Support:** Supported students with UX methods including heuristic evaluation, cognitive walkthroughs, and usability testing, and interpreted findings into improvements.

- **UX Artefacts & Prototyping:** Guided creation of personas, user workflows, and journey maps, and supported evaluation of low-, mid-, and high-fidelity prototypes.
- **Assessment & Feedback:** Marked essays, assignments, and examinations; provided clear, criteria-based feedback.
- **Student & Course Support:** Staffed student helpdesk sessions and online discussion forums; attended staff meetings with coordinating lecturers.

University of Kelaniya
 Tutor - Computer Science
 Lecturer - Computer Science

Colombo, Sri Lanka
 Mar.2017 - Sep.2017
 Sep.2017 - May.2021

- **Teaching Delivery:** Prepared and delivered lectures; facilitated tutorials, seminars, and laboratory sessions; stimulated and guided class discussions.
- **Assessment & Learning Support:** Designed and marked essays, assignments, and examinations; advised and supported students on academic matters.
- **Curriculum & Academic Planning:** Contributed to course/degree requirements, curriculum review, and academic planning; compiled bibliographies and curated specialised reading materials.
- **Supervision & Leadership:** Supervised honours student work programs and supported/supervised tutorial staff and teaching delivery.
- **Health Data Analysis (Diary Studies):** Partnered with the Faculty of Health to analyse 6M+ diary entries, supporting evidence-based findings and reporting.

EDUCATION

Victoria University of Wellington
 PhD in Computer Science - Specialised in Human Computer Interactions
 Thesis: Role of Conversational Virtual Characters in Improving Social Interactions of Adolescents in Collaborative Games in Social VR

Wellington, New Zealand
 Mar.2021 - Jul.2025

University of Kelaniya
 B.Sc (Special) in Computer Science
 Research Project: Braille Messenger: Adaptive learning based non-visual touch screen text input for the blind community using braille

Colombo, Sri Lanka
 Mar.2013 - Mar.2017

TECHNICAL SKILLS

Research Methods	Interviews, focus groups, surveys/questionnaires (Qualtrics), observations, Telemetry
Evaluation Methods	Usability testing, heuristic evaluation, cognitive walkthrough, prototype evaluation (low/high fidelity), User Experience, Playtestings
Study & Experiment Design	Within-subject, between-subjects, mixed factorial (within + between), partially nested designs, A/B testing
Visual Tools for Insights and Decision Making	Affinity diagramming, sociograms (multi-human-AI interaction behaviour analysis), Priority matrix, impact vs, effort diagrams,
UX Tools & Frameworks	Personas, user interaction workflows, journey maps, wireframing

Data Analysis	Quantitative data analysis, descriptive & inferential statistics, data visualisation, Tableau, R, Python, SQL, Qualitative data analysis, thematic analysis, content analysis
Research Tools	NVivo, Qualtrics, Miro, RStudio
AI/ML Platforms & Services	OpenAI, Huggingface, Kaggle, Azure Cognitive, Copilot, Cursor.ai
Prototyping	Low/high-fidelity prototyping, pen & paper, Figma, ShapeXR, Unity functional prototypes, XR/VR
Software & databases	Unity, .Net, C#, C, Java, HTML, CSS, JS, D3.js, Bootstrap, React, PHP, MySQL, PL/SQL, Google Colab, Jira, Git & Git Projects

FEATURED PROJECTS

Social Buddy: AI-Powered NPC for Social Virtual Reality

- Developed a conversational virtual agent in Social VR to support engagement for adolescents with social interaction difficulties based on insights from mixed method approaches in Human-centered designs.
- Comprehensive evaluation study to validate the designed mechanics.

Tech Stack: Unity3D, Unity Services, C#, OpenAI, Azure Services, Git

Mid-Air Hand Gesture Video Controller

- Fine-tuned a Vision Transformer (ViT) model to recognise custom mid-air hand gestures for video player control.
- Integrated the model into a Unity prototype for real-time inference and interaction.

Tech Stack: Unity, C#, Python, Hugging Face, Computer Vision/ML, Git

Blind Low Vision New Zealand - See Through

- Conducted Meeting with Stakeholders as a part of Design Thinking process where prioritised empathizing and identifying requirements.
- Built two prototypes to simulate various visual impairments to meet their requirements.

Tech Stack: C#, Unity 3D, Shader Graphs

HONOURS & AWARDS

- **Appreciation Letter for Outstanding Performance - HCI - Victoria University of Wellington** 2021
Received an appreciation letter from the Head of School for receiving an A+ for HCI course.
- **Batch Top - General Convocation 2017 - University of Kelaniya** 2017
Memorial gold medal for receiving the highest marks of the batch. GPA: 3.96/4.00
- **Computer Science Gold Medal - General Convocation 2017 - University of Kelaniya** 2017
Gold medal for receiving the highest marks for Computer Science. SGPA: 4.0/4.0
- **Best Undergraduate Research Project - Computer Science Category - Sri Lanka** 2017
Sri Lanka Association for the Advancement of Science (SLAAS)
- **Dean List - Academic Excellence - University of Kelaniya** 2015/2016
Nominated for the Dean's List, Faculty of Science. Yearly (2 semesters) YGPA: 4.0/4.0